Technological University of the Philippines Visayas
COLLEGE OF ENGINEERING
Talisay City, Negros Occidental

SRAM Implementation with CPLD

A Project presented to the

Computer Engineering Department

of Technological University of the Philippines Visayas

In Partial Fulfillment

of the Requirements for the Subject

HDL

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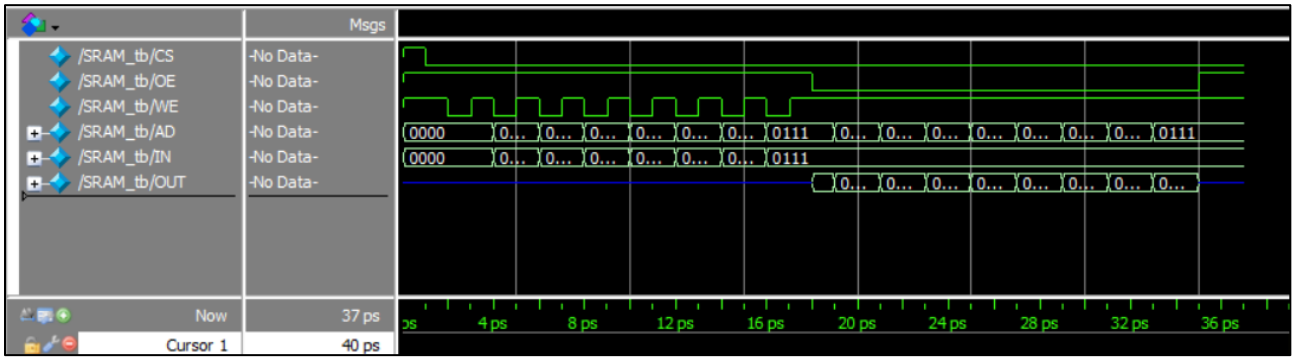
MODELSIM SIMULATION

Verilog Code Testbench

```
module SRAM_tb;

        reg CS, OE, WE;
        reg [3:0] AD, IN;
        wire [3:0] OUT;

        SRAM U1(OUT,CS,OE,WE,AD,IN);

        initial begin
                CS = 1;
                OE = 1;
                WE = 1;
                AD = 4'b0000;
                IN = 4'b0000;
        end

        initial begin
                #1 CS = 0;

                #1 AD = 4'b0000; IN = 4'b0000; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0001; IN = 4'b0001; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0010; IN = 4'b0010; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0011; IN = 4'b0011; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0100; IN = 4'b0100; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0101; IN = 4'b0101; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0110; IN = 4'b0110; WE = 0;
                #1 WE = 1;
                #1 AD = 4'b0111; IN = 4'b0111; WE = 0;
                #1 WE = 1;

                #1 OE = 0;
                #1 AD = 4'b0000;
                #1 $display("The data at address %b is %b", AD, OUT);
                #1 AD = 4'b0001;
                #1 $display("The data at address %b is %b", AD, OUT);
```

```
                #1 AD = 4'b0010;
                #1 $display("The data at address %b is %b", AD, OUT);
                #1 AD = 4'b0011;
                #1 $display("The data at address %b is %b", AD, OUT);
                #1 AD = 4'b0100;
                #1 $display("The data at address %b is %b", AD, OUT);
                #1 AD = 4'b0101;
                #1 $display("The data at address %b is %b", AD, OUT);
                #1 AD = 4'b0110;
                #1 $display("The data at address %b is %b", AD, OUT);
                #1 AD = 4'b0111;
                #1 $display("The data at address %b is %b", AD, OUT);

                #1 OE = 1;
                #1 $display("The data at AD %b is %b", AD, OUT);

                #1 $finish;
        end
endmodule
```

Timing Diagram

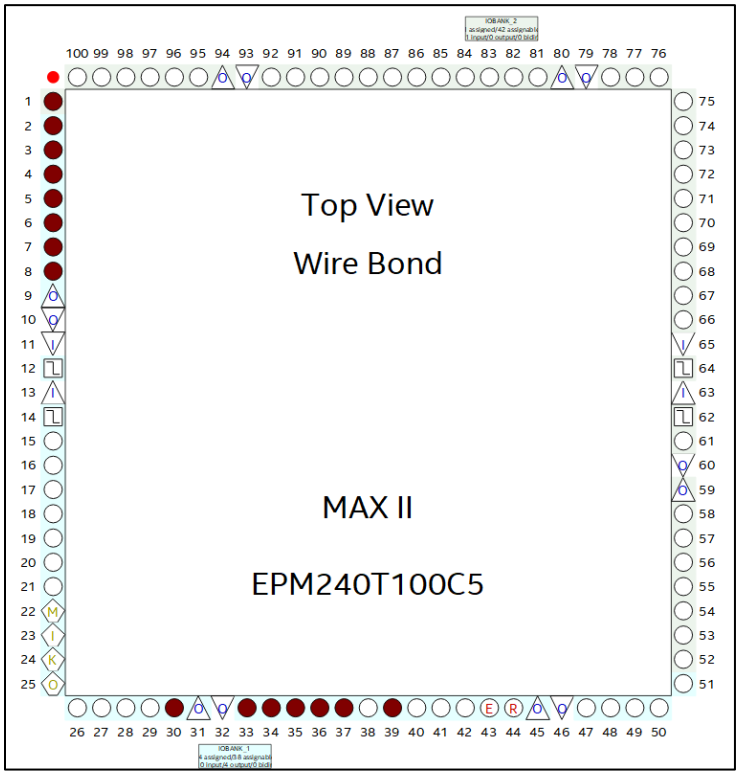


QUARTUS

Verilog Code

```
module SRAM_Design (OUT,CS,OE,WE,AD,IN);

   output [3:0] OUT;
   input CS, OE, WE;
   input [3:0] AD, IN;
   reg [3:0] mem[0:15];

   integer i;

   initial begin
         for (i = 0; i < 16; i = i + 1)
               mem[i] = 4'b0000;
   end

   assign OUT = (CS == 1 | OE == 1 | WE == 0) ? 4'bZZZZ : mem[AD];

   always @(negedge WE) begin
         if(!CS) mem[AD] <= IN;
   end

endmodule
```

Pin Planner



Node Name	Direction	Location	I/O Bank	itter Locatior	I/O Standard	Reserved	urrent Streng	ict Preservati
AD[3]	Input	PIN_1	2	PIN_3	3.3-V ...fault)		16mA ...ault)	
AD[2]	Input	PIN_3	1	PIN_67	3.3-V ...fault)		16mA ...ault)	
AD[1]	Input	PIN_5	1	PIN_16	3.3-V ...fault)		16mA ...ault)	
AD[0]	Input	PIN_7	1	PIN_56	3.3-V ...fault)		16mA ...ault)	
CS	Input	PIN_30	1	PIN_66	3.3-V ...fault)		16mA ...ault)	
IN[3]	Input	PIN_2	1	PIN_58	3.3-V ...fault)		16mA ...ault)	
IN[2]	Input	PIN_4	1	PIN_42	3.3-V ...fault)		16mA ...ault)	
IN[1]	Input	PIN_6	1	PIN_17	3.3-V ...fault)		16mA ...ault)	
IN[0]	Input	PIN_8	1	PIN_4	3.3-V ...fault)		16mA ...ault)	
OE	Input	PIN_34	1	PIN_90	3.3-V ...fault)		16mA ...ault)	
OUT[3]	Output	PIN_33	1	PIN_62	3.3-V ...fault)		16mA ...ault)	
OUT[2]	Output	PIN_35	1	PIN_57	3.3-V ...fault)		16mA ...ault)	
OUT[1]	Output	PIN_37	1	PIN_53	3.3-V ...fault)		16mA ...ault)	
OUT[0]	Output	PIN_39	1	PIN_40	3.3-V ...fault)		16mA ...ault)	
WE	Input	PIN_36	1	PIN_14	3.3-V ...fault)		16mA ...ault)	
<<new node>>								

Flow Summary

Flow Summary

<<Filter>>

Flow Status	Successful - Sat Jan 25 23:31:19 2025
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	SRAM_Design
Top-level Entity Name	SRAM_Design
Family	MAX II
Device	EPM240T100C5
Timing Models	Final
Total logic elements	105 / 240 (44 %)
Total pins	15 / 80 (19 %)
Total virtual pins	0
UFM blocks	0 / 1 (0 %)